

Chapter 1: Introduction

Digital Education (CS-411)

September 8, 2026

Team

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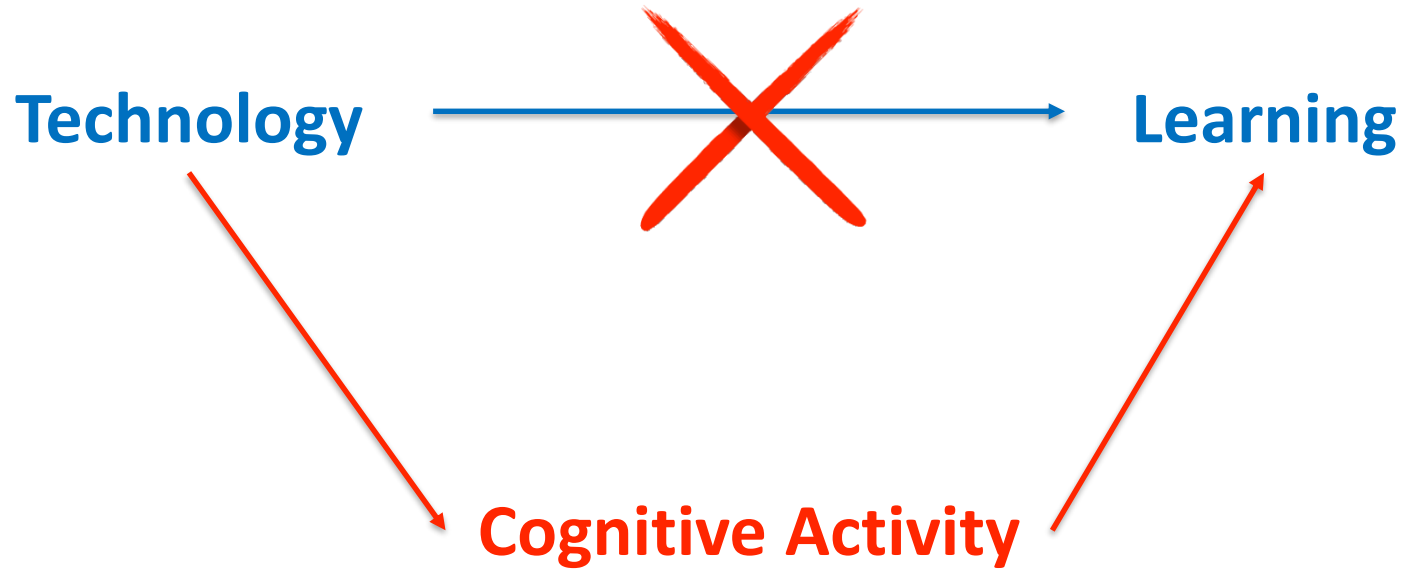
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Students – Shake Hands



Today

- What is this course about?
- Course logistics
- Introduction to project
 - Project guidelines
 - Explorative activities



what learners learn does not depend on the technology, but on the cognitive activity they have to do (with the technology)



CS-411



Drill & Practice

Intelligent tutoring systems

Online courses

Learning management systems

Classroom participation systems

Simulations

Serious Games

Microworlds

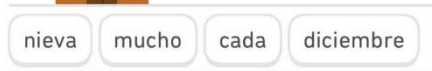
Augmented reality

Virtual reality

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Translate this sentence



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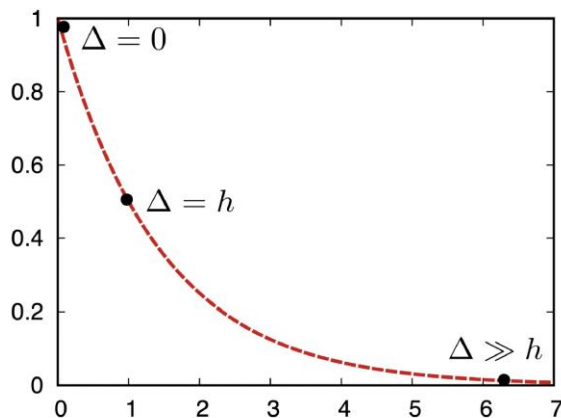
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$$p = 2^{-\left(\frac{\Delta}{\bar{h}}\right)}$$

p = Probability of recalling an item

Δ = Time elapsed since the item was reviewed

$\bar{h}$ = Half-life of the item

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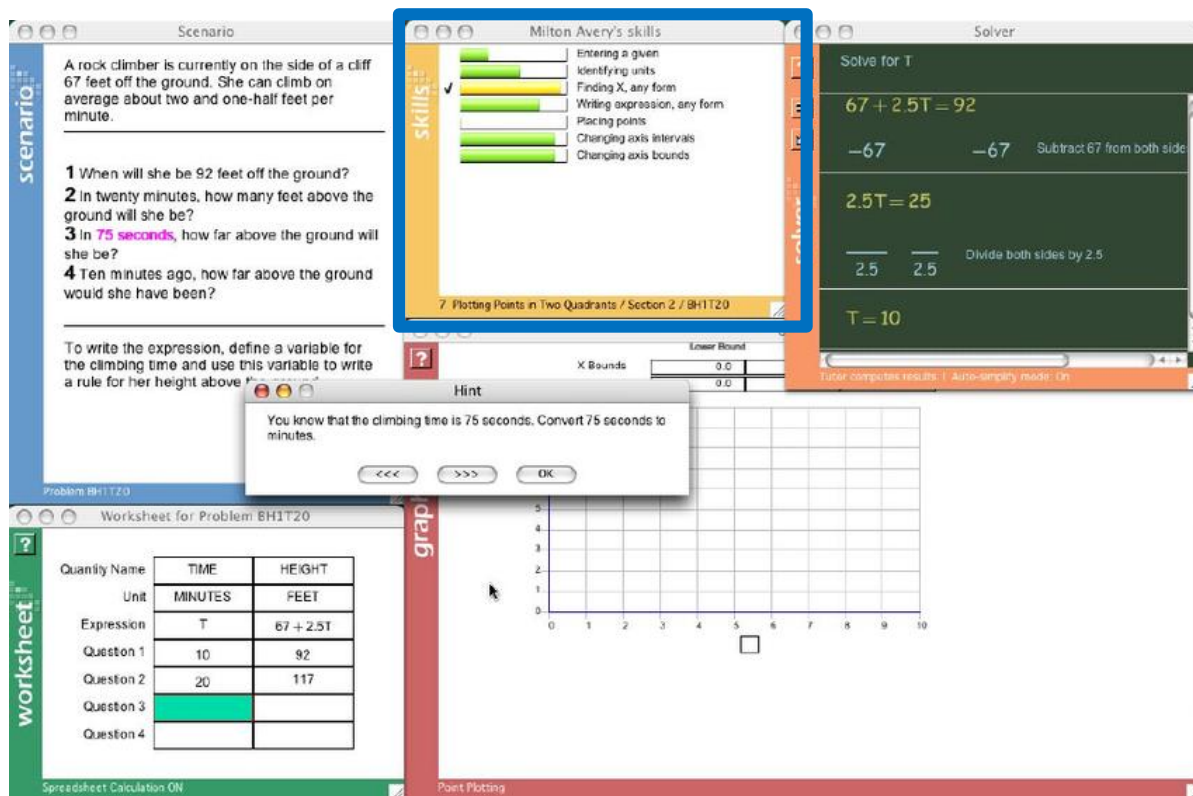
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Skill estimation



Anderson, J. R., Corbett, A. T., Koedinger, K. R., & Pelletier, R. (1995). Cognitive Tutors: Lessons Learned. *Journal of the Learning Sciences*, 4(2), 167–207.

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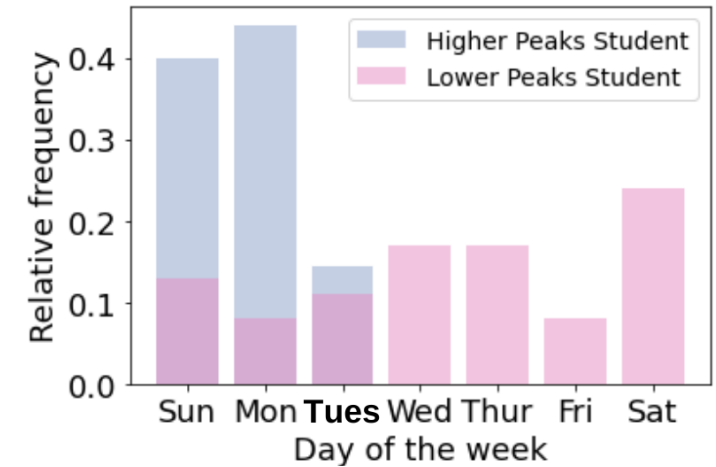
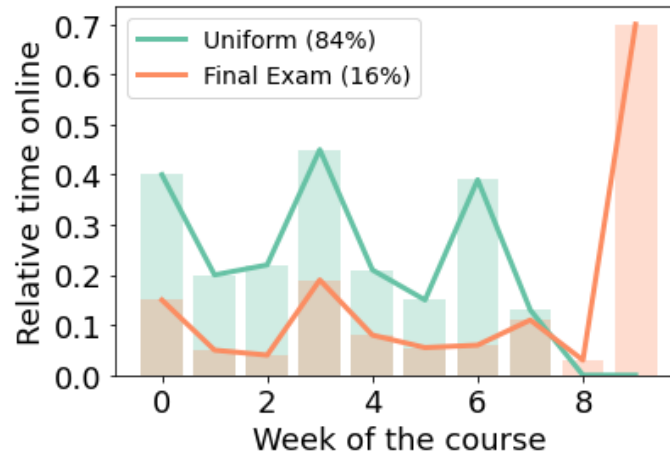
Serious Games

Microworlds

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The screenshot displays the EPFL Digital education platform interface. The top navigation bar includes links for Home, Dashboard, My courses, and My Media. The main content area is titled "Digital education" and shows course information for "Computer Science (IN) / IN - Master". A sidebar on the left lists various courses (CS 101, ECON 102, MATH 201, ENGG 202) and categories (General, Lectures, Tutorials, Problem Sets, Assignments, Midterm, Exam). The main content area displays a discussion thread titled "Quadratic equation" with a pinned message "Welcome!" and a post by "Anonymous" asking for help solving a quadratic equation. The post has 4 replies and 242 views. Below the post, there is an answer by "Scott Maxwell" (STAFF) providing the quadratic formula:
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
 and a graph of a parabola. The graph shows a blue parabola opening upwards on a coordinate plane with x and y axes ranging from -4 to 4. The vertex is at (0, 0). The x-axis is labeled from -4 to 4, and the y-axis is labeled from 0 to 20. The parabola passes through points (-2, 4), (-1, 1), (0, 0), (1, 1), and (2, 4).

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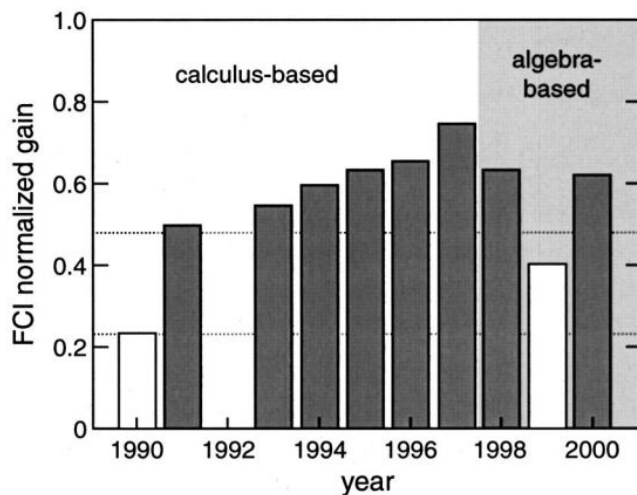
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Crouch, C.H., & Mazur, E. (2001). Peer Instruction: Ten years of experience and results. *American Journal of Physics*, 69, 970-977.

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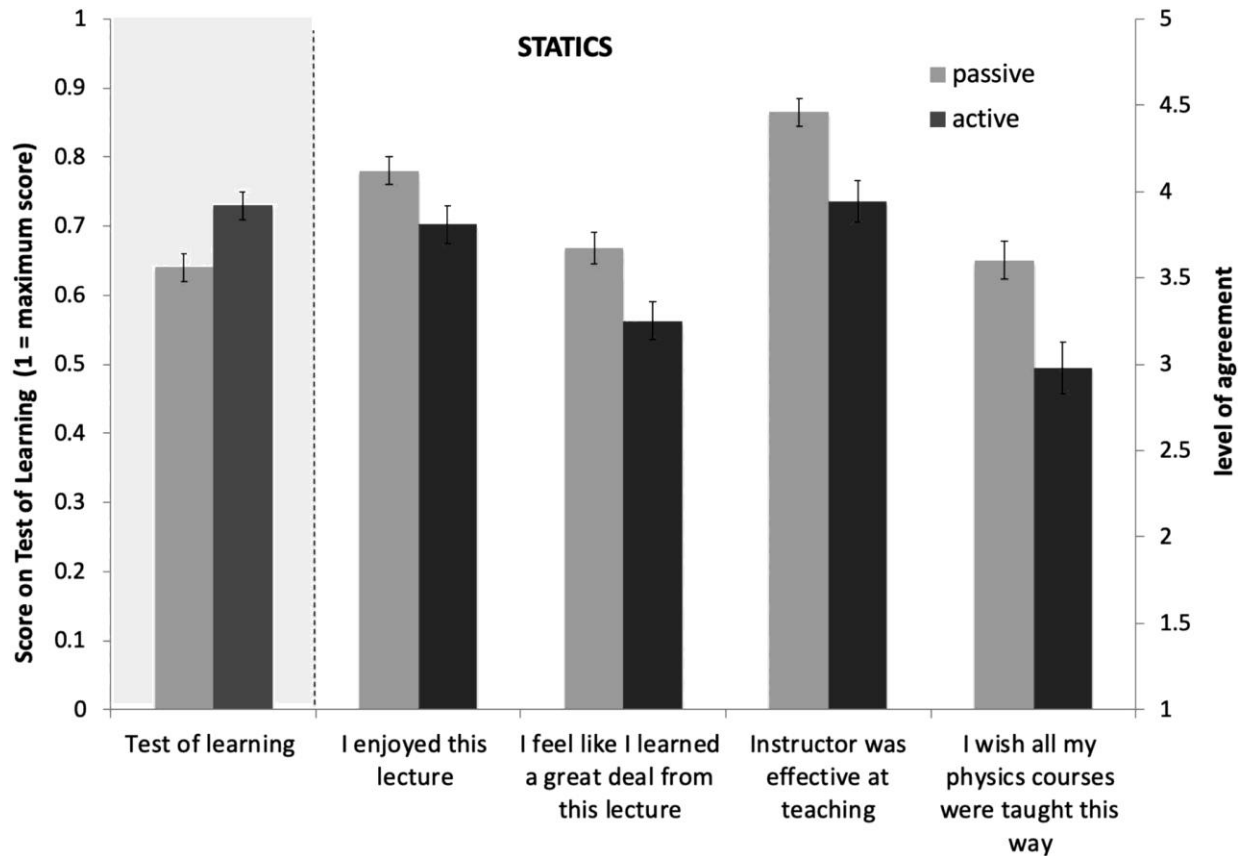
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Deslauriers, L., McCarty, L. S., Miller, K., Callaghan, K., & Kestin, G. (2019). Measuring actual learning versus feeling of learning in response to being actively engaged in the classroom. *PNAS*, 116(39), 19251–19257.

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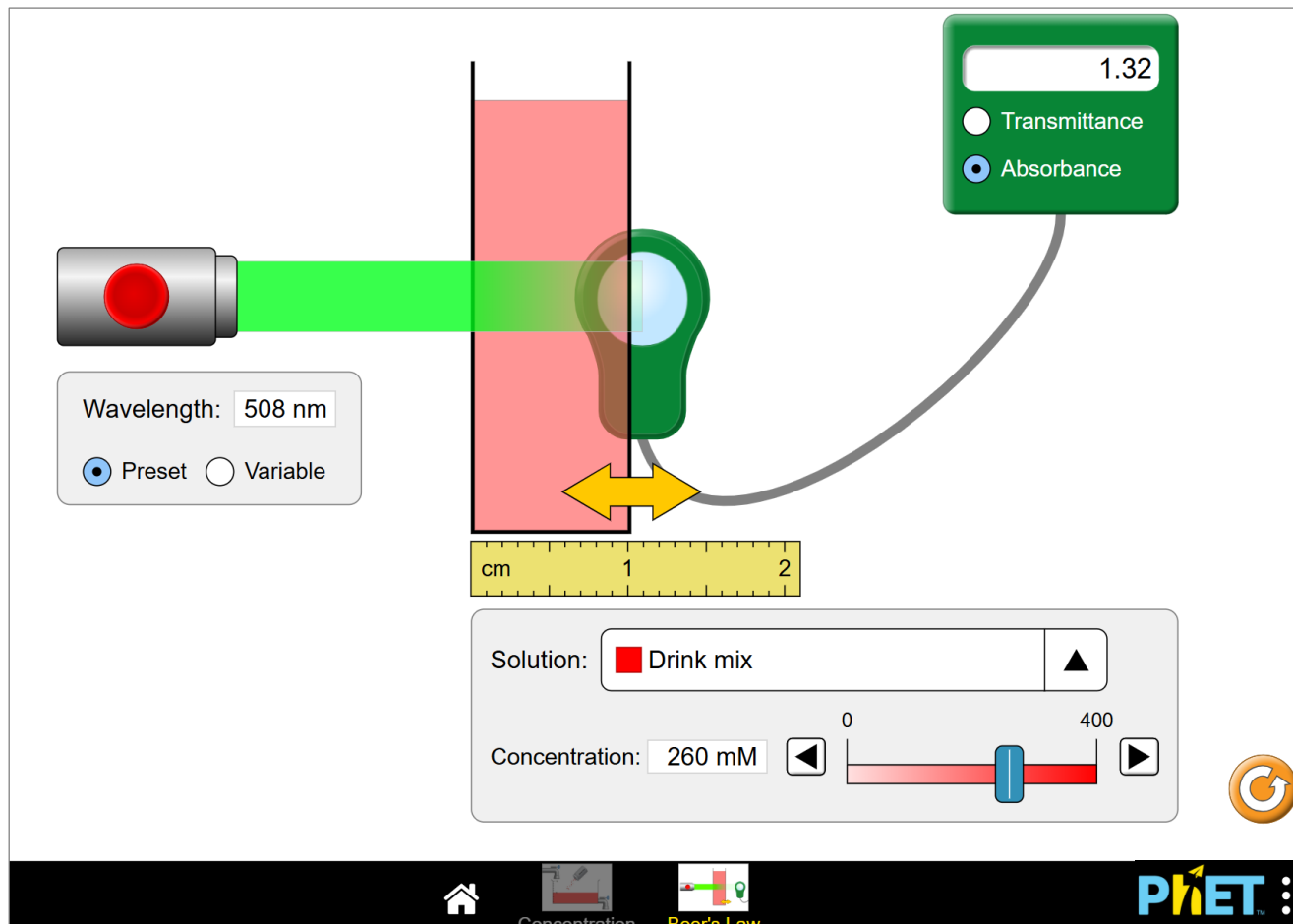
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The image shows the PhET Beer's Law simulation interface. A green light beam from a source on the left passes through a pink rectangular container. A green detector on the right measures the light. A yellow double-headed arrow indicates the path of the light. A green box in the top right corner displays the value 1.32, with radio buttons for Transmittance (unselected) and Absorbance (selected). A grey box on the left shows Wavelength: 508 nm, with radio buttons for Preset (selected) and Variable. A yellow ruler below the container shows a scale from 0 to 2 cm. A grey box at the bottom shows Solution: Drink mix (selected) and Concentration: 260 mM, with a slider ranging from 0 to 400 mM. A PhET logo is in the bottom right corner.

Wavelength: 508 nm

☒ Preset ☐ Variable

1.32

☐ Transmittance ☒ Absorbance

cm 1 2

Solution: Drink mix

Concentration: 260 mM

0 400

PhET

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1 Patient's Symptoms

Symptom A	Symptom B
Pain	Stomach Cramps
Symptom C	Symptom D
Vomiting	No entry

2 Test Results

I have tested:	The results showed:
Object A Egg	Not contaminated
Object B Orange	Not contaminated
Object C Cheese	Not contaminated
Object D Water	Not contaminated

3 Possible Explanations

	Likelihood:	Because:
Anthrax	Unlikely	Characteristics don't match
Botulism	No entry	No entry
Ebola	No entry	No entry
Influenza	Unlikely	Characteristics don't match
Salmonellosis	Possible	No entry
Smallpox	Very unlikely	Characteristics don't match

4 Final Diagnosis

Disease
No entry

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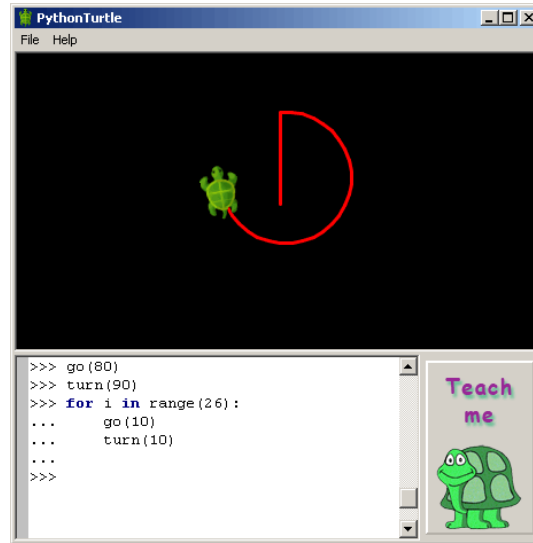
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<https://turtleacademy.com/>



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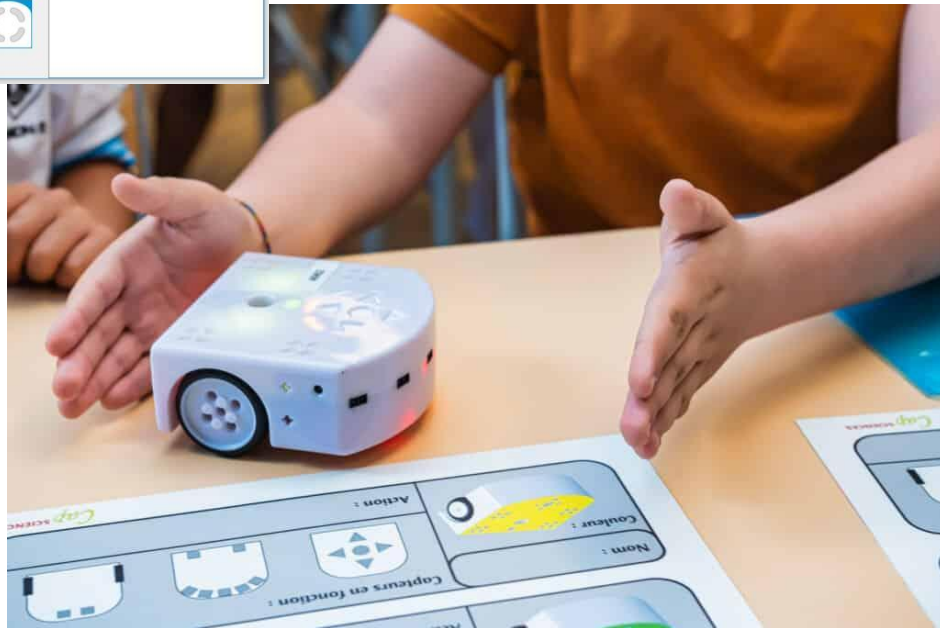
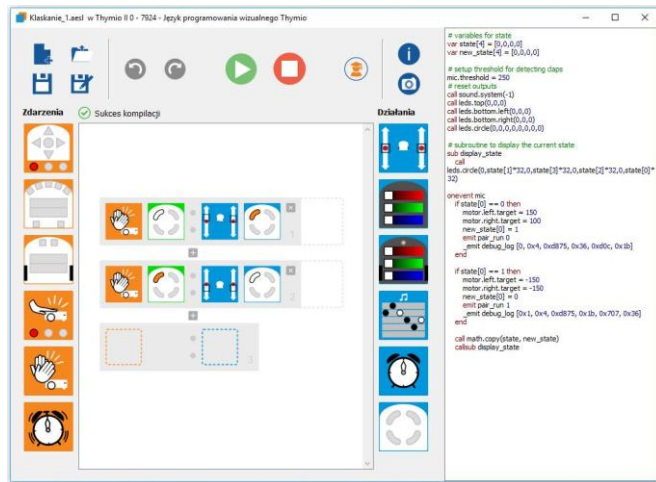
Serious Games

Microworlds

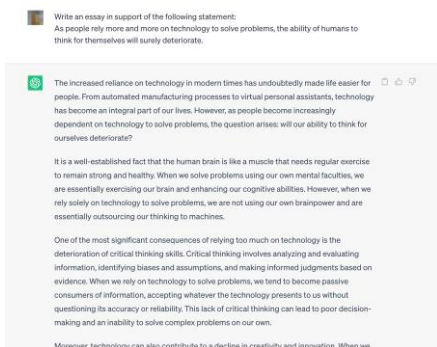
Augmented reality

Virtual reality

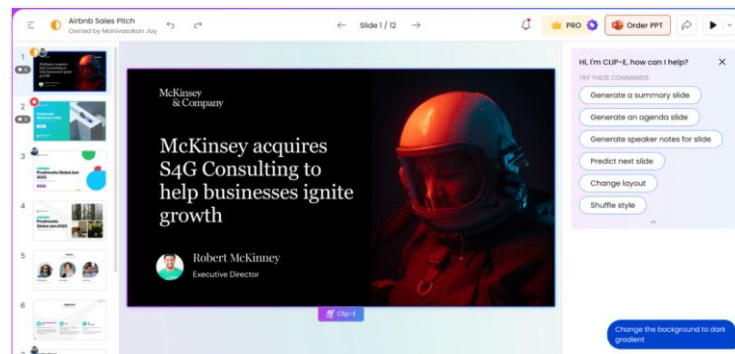
Educational Robotics



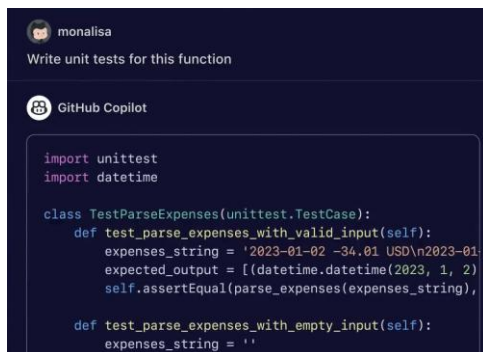
What about generative AI?



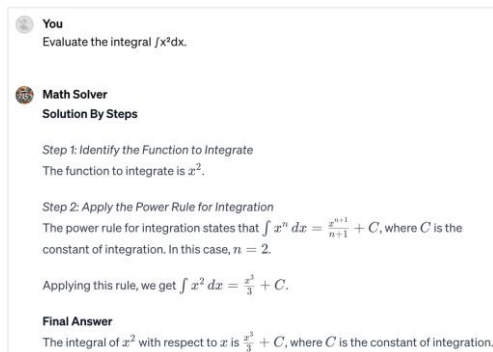
Writing



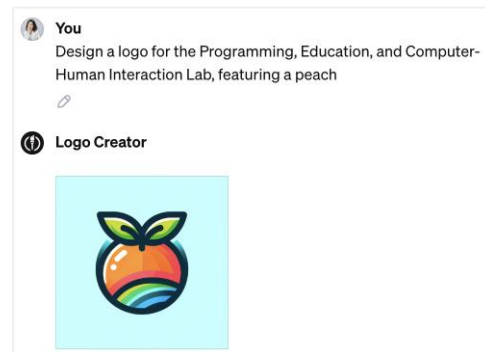
Presentation



Programming



Math



Design

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AI Chatbots

 ChatGPT   



I don't understand why we divide by $n - 1$ instead of n when calculating the sample variance. Can you explain?



Sure! The reason is that when we calculate the sample variance, we first estimate the population mean using the sample mean. This uses up one degree of freedom.

Imagine you have three observations with a known sample mean. Once you know the first two observations, the third is constrained: it has to take a particular value for the mean to stay the same. So only $n - 1$ deviations from the sample mean can vary freely.

Dividing by $n - 1$, rather than n , corrects for the fact that the sample variance would otherwise tend to underestimate the population variance. This correction is called Bessel's correction.

Would you like me to show this with a small numerical example?



Yes, please.



Great! Suppose our sample is 2, 4, 6. The sample mean is 4. The squared deviations are:

$$(2 - 4)^2 = 4, \quad (4 - 4)^2 = 0, \quad (6 - 4)^2 = 4$$

Their sum is 8. If we divide by $n = 3$, we get $8/3 \approx 2.67$. If we divide by $n - 1 = 2$, we get 4, which is the unbiased estimate of the population variance.

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That's alright, Rose. 😊 Ratios can be a bit tricky. Let's break it down. The explanation suggests I'll use here will be focusing on the horizontal squares first. Can you see how many horizontal squares there are between point P and point Q? 📐

LearnLM

Tutor

Student 4 😊

Spot on! 🎉 And how many horizontal squares are there between point Q and point R? 😊

LearnLM

Student 8 🙌

Excellent! Now, what's the ratio of 4 to 8? Can you simplify it?

LearnLM

Student

<https://arxiv.org/html/2512.23633v1>

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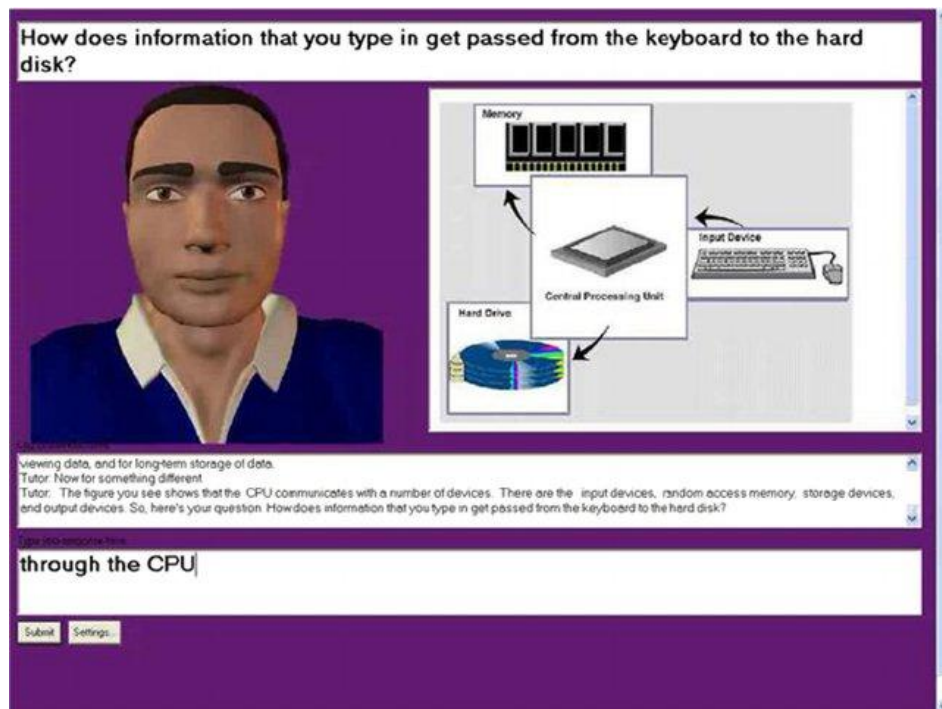
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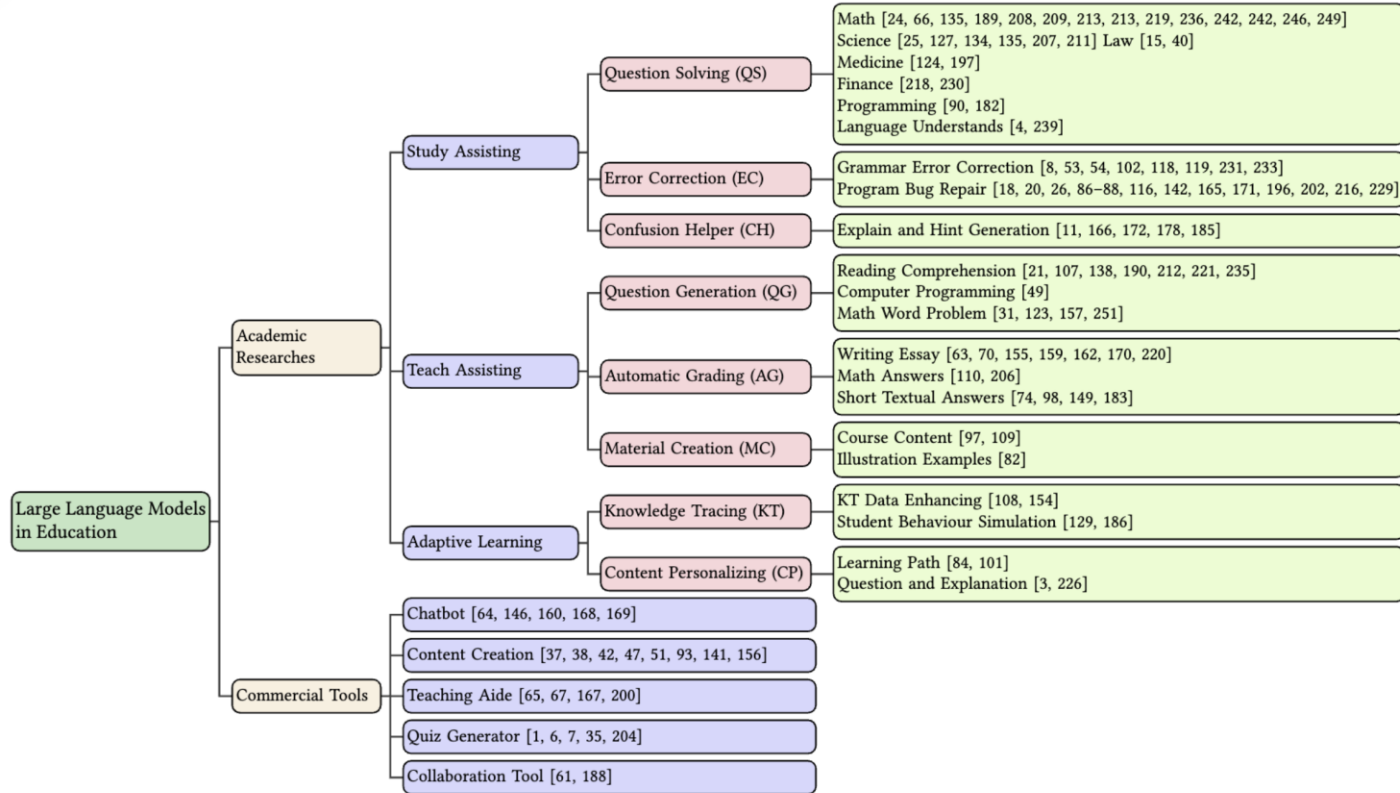
AI Chatbots



GenAI enables flexible, open-ended natural language interaction.

Graesser, A. C., Lu, S., Tanner Jackson, G., Mitchell, M. M., Ventura, M., Olney, A. & Louwerse, M. M. (2004). AutoTutor: A tutor with dialogue in natural language. *Behavior Research Methods, Instruments, & Computers* 36, 180–192

LLMs – One model for it all!



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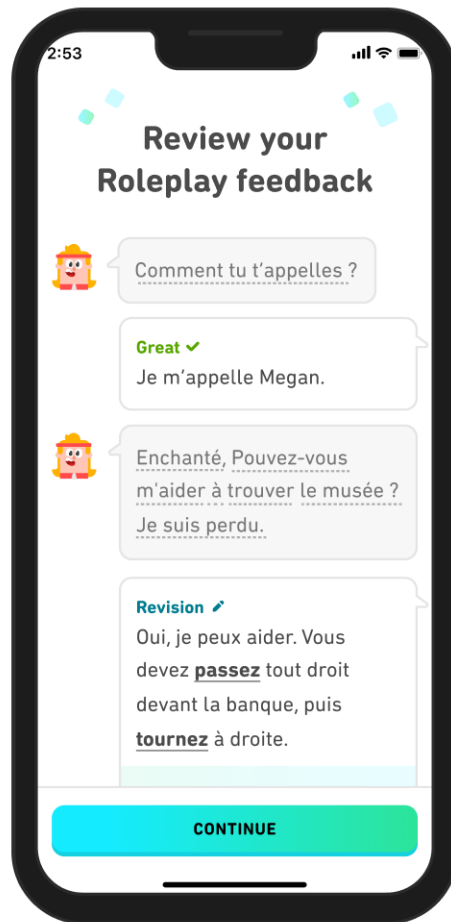
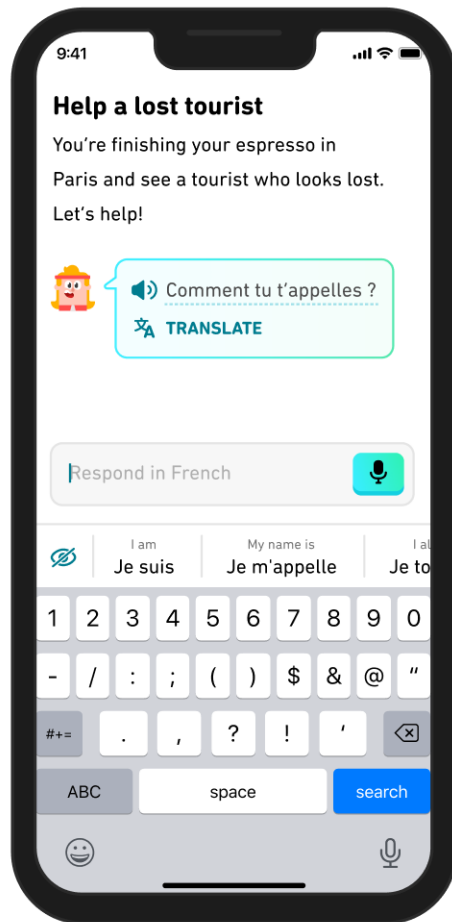
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Simulation

Great! Then we'll get started right away.

What happens to the absorbance when the width doubles? Please write down your hypothesis.

I don't know. Maybe it will double in size?

Thank you — I've noted your suggestion.

Try this out in the simulation: First, set a width, read the absorbance, then double the width and read the new absorbance. Afterwards, please send me both values (width, concentration, absorbance) for both states.

(1, 340, 1.72)

Great, that's a good value! Now I need a second.

Ask Matteo...

Send

COLLECT DATA

PhET

E. Shved, E. Bumbacher, S. P. Neshaei and T. Käser (2026). Teaching Multivariational Reasoning through AI-Guided Inquiry in Interactive Simulations, AIED.

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AI Chatbots

many technologies!

how do they support learning?

how do we learn?

how can we design AI affordances that
foster learning?

Week	Lecture (10:15-12:00)	Lab (08:15-10:00)
1	Introduction + Project Presentation	No lab session
2	Human Cognition	Defining learning objectives
3	Piaget, Microworlds, Simulations	Designing learning activities
4	Skinner, Bloom, and Mastery Learning	Designing learning activities
5	GenAI in Education I	Implementing learning activities
6	Social Aspects of Learning	Implementing learning activities
7	Autumn Break	Autumn Break
8	GenAI in Education II	Implementing learning activities
9	Designing Experiments	Running experiments
10	Statistics 101	Statistics
11	Qualitative Methods	Running experiments
12	Reserve	Running experiments
13	No Lecture	Analyzing data
14	No Lecture	Analyzing data
15	Project Presentations	Project Presentations

How do we learn?

What affordances of
AI can support
learning?

How can we
measure learning?

CS-411: 6 credits

- Written exam (50%): we will make a past example available on Moodle
- Project (50%):
 - Teams of 3 people
 - Select a topic that is difficult to understand
 - Design learning activities with and without AI affordance
 - Run an experiment and evaluate it
 - There will be intermediate **mandatory milestones**
 - **Present the final project on December 15 (in-person attendance mandatory)**
 - 20% initial draft (M3), 20% plan + working conditions (M4), 30% poster presentation (M6), 30% final report (M7)

Detailed instructions on Moodle

Week	Lab (08:15-10:00)	Project Milestones	Grade
1	No lab session		
2	Defining learning objectives	C1: Thursday, September 17, 23:59 Submit: group formation, topic, initial reflections on GenAI use for learning	Mandatory, ungraded
3	Designing learning activities	C2: Thursday, September 14, 23:59 Submit 1-2 pages description of learning goals and learning activity	Optional, ungraded
4	Designing learning activities		
5	Implementing learning activities	M3: Thursday, October 8, 23:59 Submit 3-4 pages project plan, first working prototype of AI-supported activity	20% of project grade
6	Q&A + Feedback with TAs (graded)	Mandatory sessions, Zoom attendance possible	
7	Autumn Break		
8	Implementing learning activities		
9	Running experiments	M4: Thursday, November 5, 23:59 Submit 2-3 pages study plan, completed implementations for both conditions	20% of project grade
10	Q&A + Feedback with TAs (graded)	Mandatory sessions, Zoom attendance possible	
11	Running experiments		
12	Running experiments	C5: Thursday, November 26, 23:59 Submit 1-2 pages initial results (or planned analyses)	Optional, ungraded
13	Analyzing data		
14	Analyzing data		
15	Project Presentations	M5: Tuesday, December 15, 9:00-12:00 Mandatory in-person presentations followed by Q&A	30% of project grade
	Final Report	M6: Thursday, January 7; 23:59 Submit final report	30% of project grade

Detailed instructions on Moodle

The project

- Design a lesson with and without AI affordances
- We expect you to **use generative AI as part of your project work**, including to support coding and other project tasks.
 - You remain responsible for all content and work you submit.
 - You must be able to justify your design decisions and explain all choices, analyses, and results.
- **Document your interactions with generative AI.** We will use these interactions throughout the semester to reflect on how AI can support—or hinder—learning and work, connecting your own experiences to the theories discussed in class.

The objectives

- You will learn to:
 - Define specific, measurable learning goals
 - Apply learning theories to design activities
 - Implement an AI-supported digital lesson
 - Design and execute an experiment comparing AI-support with a non-AI version.
 - Analyze data quantitatively and qualitatively
 - **Collaborate effectively with AI**
 - **Analyze how different ways of using AI can support or hinder learning**
 - **Critically reflect on your own use of AI for learning**

Your Experience

- ① How are you using generative AI for your learning?
- ② What tools are you using and how are you using them?
- ③ How do they support your learning?

Three Examples

- REFINE – AI feedback
- PharmaSim – AI scaffolding
- Pensée – AI-supported reflection

Analyze the learning activity

- ① Where and how is AI used in the learning environment?
- ② What learning activity is the AI affordance intended to support?
- ③ How might this affordance support—or hinder—learning?

Pensée – Supporting reflection

1

Reflecting on what you did well during your time with the patient. What aspects of the situation did you find easy, and why do you think that was? Feel free to describe your experience in detail!

3:46 PM

I think the easy thing was that the patient was very cooperative

3:46 PM

It's great that the patient was cooperative, as that can make the process smoother. Now, let's think about the challenges you faced. Were there any difficulties or mistakes you encountered during the session? What made them

Stage 1 of 4

Type your message here...

Send

Key Concepts

Family interruptions during session

the family of the patient was very impatient and interrupted us several times by trying to come to the room.

Patient was very cooperative

I think the easy thing was that the patient was very cooperative

2

During my hospital visit, I was mainly involved in communicating with patients and observing the work of healthcare staff. I introduced myself, followed hygiene protocols, and tried to support patients through conversation and attentive listening. At first, I felt confident and motivated, but I also felt emotionally affected by seeing people in distress. Overall, the experience was positive, but I noticed some weaknesses, such as giving overly detailed explanations and not immediately recognizing when a patient was tired or uncomfortable. This made some interactions less effective than they could have been.

Get Feedback

Gibbs Reflective Cycle

```
graph TD; Description[Description] --> Feelings[Feelings]; Feelings --> Evaluation[Evaluation]; Evaluation --> Analysis[Analysis]; Analysis --> Conclusion[Conclusion]; Conclusion --> ActionPlan[Action Plan]; ActionPlan --> Description;
```

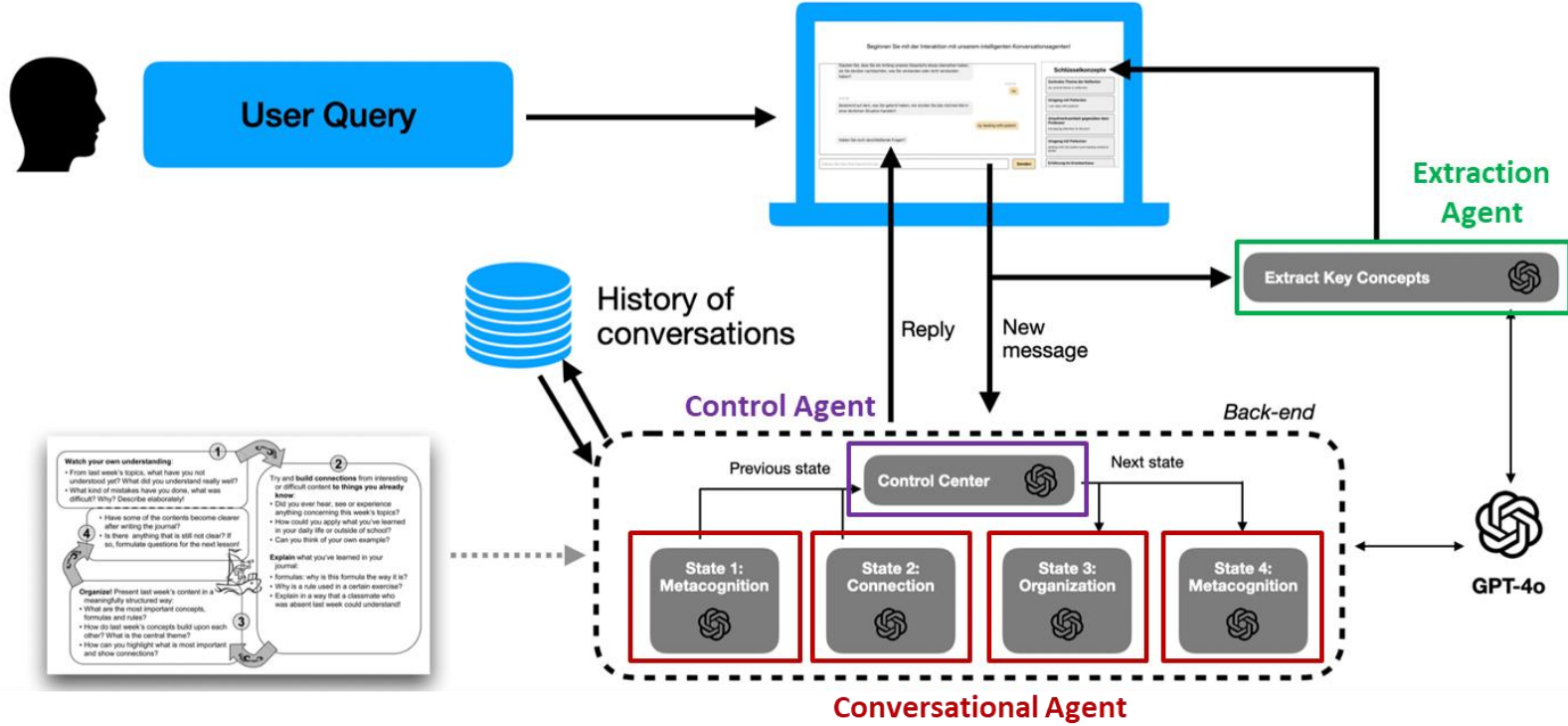
This class is used to describe how you felt during the situation.

Examples of Feelings:

- I felt anxious and overwhelmed when I realized that the patient's condition was deteriorating. However, I was also happy because I had studied the symptoms and knew how to respond.

Neshaei, S. P., Davis, R. L., & Käser, T. (2026). Moving Beyond Review: Applying Language Models to Planning and Translation in Reflection. *AIED*

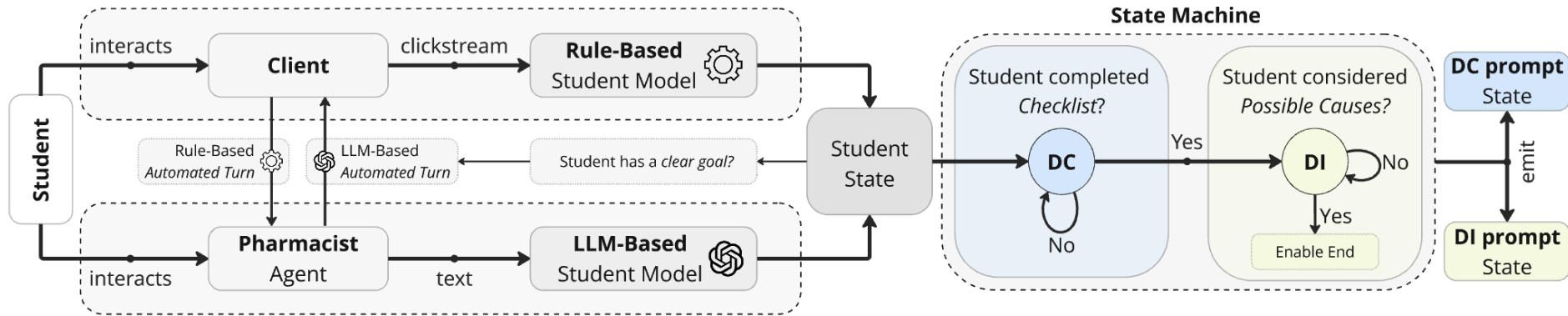
Pensée – Behind the scenes



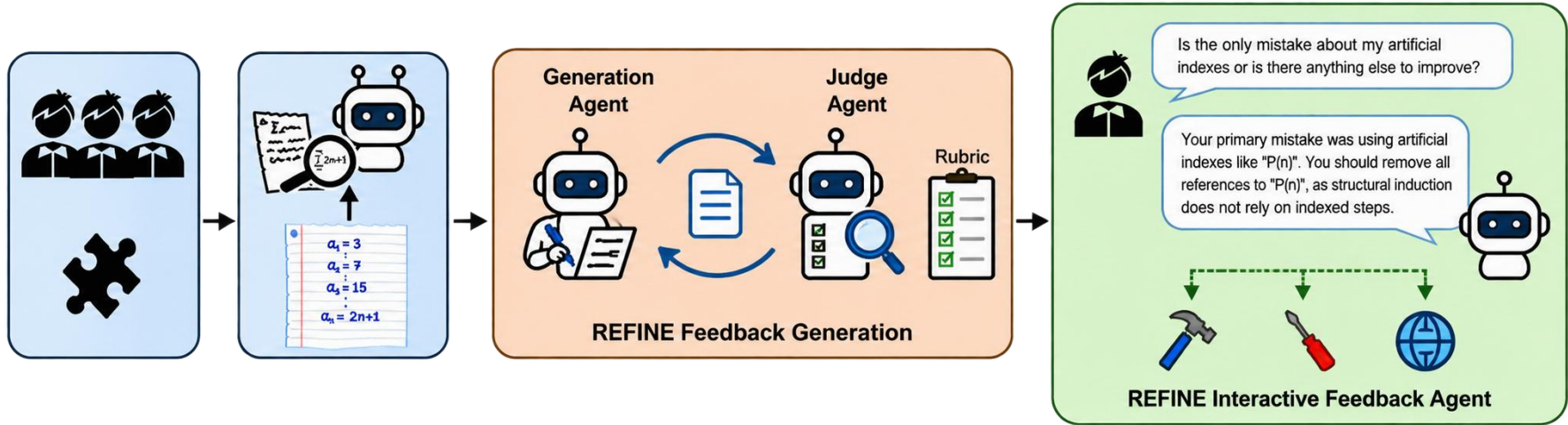
PharmaSim – Supporting diagnostic reasoning



PharmaSim – Behind the scenes



REFINE



See you next week at 8:15am!